The low-component cases for total cut complexes of disconnected graphs

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Abstract

Let $G = G_1 \sqcup \cdots \sqcup G_k$ be a graph with k nonempty connected components and n vertices. For $d \geq 2$, write $\Delta_d^t(G)$ for the total d-cut complex and $\mathrm{BI}_d(G)$ for its Alexander dual. For $d \geq 3$, Carnero Bravo proved that, under natural contractibility and simple-connectivity assumptions on the components, if $k \geq d+3$, then

$$\Delta_d^t(G) \simeq \bigvee_{\binom{k-1}{d-1}} S^{n-d-1},$$

and asked whether the same formula holds for $k = d, d+1, d+2$. We also give the elementary independent argument needed for the stated $d = 2$ high-component range.

We answer all three cases affirmatively. For $k = d+2$, a complete 1-skeleton and a fixed-component triangular fan give simple connectivity. For $k = d+1$, we separate the excess $n-k = 0, 1$ boundary cases and fill the fundamental cycles of a spanning star by explicit triangular disks. For $k = d$, we cover the total cut complex by joins indexed by weak compositions of d. The all-ones block is the required sphere, while the union of the remaining blocks deformation retracts onto its intersection with that sphere. The last argument requires only the componentwise contractibility-or-void assumption and not the additional simple-connectivity hypothesis. Together with the previously known range, this completes the formula for every $k \geq d$ under the stated hypotheses.

1 Introduction

For a graph G and an integer $d \geq 1$, the total d-cut complex $\Delta_d^t(G)$ consists of the vertex sets whose complement contains an independent set of size d [1]. Equivalently,

$$\sigma \in \Delta_d^t(G) \quad \Longleftrightarrow \quad \alpha(G - \sigma) \geq d. \tag{1.1}$$

Total cut complexes were introduced by Bayer, Denker, Jelić Milutinović, Rowlands, Sundaram, and Xue [1]. Their topology connects graph structure, Stanley–Reisner theory, Alexander duality, shellability, and discrete Morse theory.

Carnero Bravo [2] studies both $\Delta_d^t(G)$ and its Alexander dual $\mathrm{BI}_d(G)$, the bounded-independence complex. For a disconnected graph $G = G_1 \sqcup \cdots \sqcup G_k$, [2, Theorem 26] calculates the homotopy type of $\mathrm{BI}_d(G)$ under componentwise contractibility assumptions. Alexander duality then determines the homology of $\Delta_d^t(G)$. To upgrade this homology calculation to a wedge-of-spheres homotopy type, [2, Theorem 28] proves simple connectivity by observing that every three vertices form a face when $k \geq d+3$. Its proof invokes [2, Theorem 26], which is stated for $d \geq 3$; we therefore supply the short $d = 2$ argument independently in Section 7.

This leaves exactly three component counts. In [2, Question 29], it is asked whether the same formula remains true for

$$k = d, \qquad k = d+1, \qquad k = d+2.$$

The difficulty is genuinely low-dimensional. The dual calculation already fixes the target homology, but the complete-2-skeleton argument used for $k \geq d+3$ is no longer available. The three missing layers exhibit three different boundary phenomena:

1. for $k = d+2$, the 1-skeleton is complete, but some triangles may be absent;
2. for $k = d+1$, even the 1-skeleton can have missing edges and the target may have dimension 0 or 1;
3. for $k = d$, the all-singleton case is already a sphere, and the extra faces contributed by larger components must be shown not to change its homotopy type.

Our main theorem resolves all three layers.

Theorem A. Let $d \geq 2$, and let

$$G = G_1 \sqcup \cdots \sqcup G_k$$

have $k \geq d$ nonempty connected components and n vertices. Assume that

1. for every i and every $2 \leq \ell \leq d$, the complex $\Delta_\ell^t(G_i)$ is void or contractible; and
2. $\mathrm{BI}_2(G_i)$ is simply connected for every i.

Then

$$\Delta_d^t(G) \simeq \bigvee_{\binom{k-1}{d-1}} S^{n-d-1}. \tag{1.2}$$

For $d \geq 3$, the range $k \geq d+3$ is [2, Theorem 28]. The main new content is the range $k = d, d+1, d+2$; Section 7 also closes the elementary $d = 2$, $k \geq 5$ dependency separately. Moreover, when $k = d$, condition 2 is unnecessary.

The proof for $k = d$ is naturally expressed by a weak-composition cover. This is close in spirit to the composition-poset and two-factor decompositions developed in [2, Theorems 26 and 34] and to the general polyhedral-join framework in [3]. Those tools are prior work. The point here is the particular multi-filter cover and the comparison between its nonbaseline union and its trace on the baseline sphere.

2 Definitions and conventions

All graphs are finite and simple. All simplicial complexes are finite. We distinguish the void complex, which has no faces, from $\{\varnothing\}$. A contractible complex is in particular nonvoid.

For a graph H and $r \geq 1$, set

$$\Delta_r^t(H) = \{\sigma \subseteq V(H) : \alpha(H - \sigma) \geq r\}. \tag{2.1}$$

The bounded-independence complex is

$$\mathrm{BI}_r(H) = \{\sigma \subseteq V(H) : \alpha(H[\sigma]) < r\}. \tag{2.2}$$

These two complexes are Alexander dual on the common vertex set [2]. We use the standard join $K * L$ for complexes on disjoint vertex sets.

For the weak-composition argument, it is convenient to extend the total-cut filtration by

$$\Delta_0^t(H) = \Delta^{V(H)}, \qquad \Delta_1^t(H) = \partial\Delta^{V(H)}. \tag{2.3}$$

The second identity follows because $H - \sigma$ contains an independent vertex exactly when $\sigma \neq V(H)$. If H has one vertex, then $\partial\Delta^{V(H)} = \{\varnothing\}$, viewed as S^{-1} in join formulas.

We repeatedly use the following elementary observation.

Lemma 2.1. *If deleting a vertex set σ leaves at least d nonempty connected components of G, then $\sigma \in \Delta_d^t(G)$.*

Proof. Choose one remaining vertex from each of d components. Vertices in distinct components are pairwise nonadjacent, so they form an independent d-set. □

3 The inherited homology calculation

The component hypotheses first determine the homology needed in the two connectivity arguments.

Proposition 3.1. *Under the hypotheses of Theorem A, $\Delta_d^t(G)$ has only one nonzero reduced homology group, namely*

$$\widetilde{H}_{n-d-1}(\Delta_d^t(G); \mathbb{Z}) \cong \mathbb{Z}^{\binom{k-1}{d-1}}. \tag{3.1}$$

Proof. Suppose first that $d \geq 3$. The proof of [2, Theorem 28], using its connectivity result for bounded-independence complexes, shows that the component hypotheses make every $\mathrm{BI}_\ell(G_i)$ contractible for $2 \leq \ell \leq d$. Then [2, Theorem 26] gives

$$\mathrm{BI}_d(G) \simeq \bigvee_{\binom{k-1}{d-1}} S^{d-2}. \tag{3.2}$$

Combinatorial Alexander duality gives (3.1).

For $d = 2$, $\mathrm{BI}_2(G)$ is the disjoint union, in geometric realization, of the clique complexes $\mathrm{BI}_2(G_i)$. Each is contractible under the assumptions: if the dual total-cut complex is void this is immediate, and if it is contractible, Alexander duality gives acyclicity while the assumed simple connectivity permits Whitehead's theorem. Thus $\mathrm{BI}_2(G)$ is homotopy equivalent to k points. Its reduced H_0 has rank $k-1$, which is $\binom{k-1}{1}$, and Alexander duality again yields (3.1). □

For $k = d+1, d+2$, we will combine Proposition 3.1 with direct low-dimensional connectivity arguments. The $k = d$ proof is stronger and does not use this proposition.

4 The case $k = d$

We first treat the most structured boundary layer.

Theorem 4.1. *Let $d \geq 2$, and let $G = G_1 \sqcup \cdots \sqcup G_d$ have d nonempty connected components and n vertices. If $\Delta^t_\ell(G_i)$ is void or contractible for every i and $2 \leq \ell \leq d$, then*

$$\Delta^t_d(G) \simeq S^{n-d-1}. \tag{4.1}$$

Proof. Let

$$W_d = \{(r_1, \dots, r_d) \in \mathbb{Z}^d_{\geq 0} : r_1 + \cdots + r_d = d\}$$

be the weak compositions of d into d parts. For $\mathbf{r} = (r_1, \dots, r_d) \in W_d$, define

$$X_{\mathbf{r}} = \Delta^t_{r_1}(G_1) * \cdots * \Delta^t_{r_d}(G_d). \tag{4.2}$$

We claim that

$$\Delta^t_d(G) = \bigcup_{\mathbf{r} \in W_d} X_{\mathbf{r}}. \tag{4.3}$$

Indeed, let $\sigma \in \Delta^t_d(G)$ and choose an independent d-set $S \subseteq V(G) - \sigma$. Put $r_i = |S \cap V(G_i)|$. Then $\mathbf{r} \in W_d$, and the component of σ in $V(G_i)$ lies in $\Delta^t_{r_i}(G_i)$. Hence $\sigma \in X_{\mathbf{r}}$. Conversely, if $\sigma \in X_{\mathbf{r}}$, choose an independent r_i-set in each $G_i - (\sigma \cap V(G_i))$. Their union is an independent d-set in $G - \sigma$. This proves (4.3).

The unique weak composition with every coordinate equal to 1 is $\mathbf{1} = (1, \dots, 1)$. Its block is

$$B = X_{\mathbf{1}} = \partial\Delta^{V(G_1)} * \cdots * \partial\Delta^{V(G_d)} \simeq S^{n-d-1}. \tag{4.4}$$

Let $R = W_d - \{\mathbf{1}\}$, omit any void blocks, and put

$$Y = \bigcup_{\mathbf{r} \in R} X_{\mathbf{r}}, \qquad A = B \cap Y. \tag{4.5}$$

Every $\mathbf{r} \in R$ has a zero coordinate and a coordinate at least 2. The zero coordinate supplies a full-simplex factor in (4.2), so every nonvoid $X_{\mathbf{r}}$ is a cone.

Now take a finite nonempty set $T \subseteq R$ and define

$$m_i = \max_{\mathbf{r} \in T} r_i.$$

Because the total-cut filtration decreases with its index and the component vertex sets are disjoint,

$$\bigcap_{\mathbf{r} \in T} X_{\mathbf{r}} = \Delta^t_{m_1}(G_1) * \cdots * \Delta^t_{m_d}(G_d). \tag{4.6}$$

At least one m_i is at least 2. If some $m_j = 0$, the intersection is a cone. If no coordinate is zero, every nonvoid factor of index at least 2 is contractible by hypothesis. Thus every nonempty intersection in this cover of Y is contractible.

Intersecting (4.6) with B gives

$$B \cap \bigcap_{\mathbf{r} \in T} X_{\mathbf{r}} = \Delta^t_{\max(1,m_1)}(G_1) * \cdots * \Delta^t_{\max(1,m_d)}(G_d). \tag{4.7}$$

Every nonempty complex in (4.7) is contractible for the same reason. Moreover, (4.6) contains a nonempty simplex if and only if (4.7) does. Only the zero coordinates change, from a full simplex to a simplex boundary; a factor with index at least 2 is nonvoid and contractible, hence contains a nonempty face. Therefore the covers of Y and A have exactly the same nonempty-intersection pattern.

Take as a basis-like cover of Y all nonempty finite intersections of the $X_{\mathbf{r}}$. For the inclusion $p : A \hookrightarrow Y$, the inverse image of each cover element is its intersection with B. By (4.6)–(4.7), both spaces are contractible. The local-to-global theorem [2, Theorem 10] implies that p is a weak equivalence, hence a homotopy equivalence between finite CW complexes.

The inclusion $A \hookrightarrow Y$ is also a cofibration, so A is a strong deformation retract of Y. Extending this deformation by the identity on B gives

$$\Delta_d^t(G) = B \cup_A Y \simeq B \simeq S^{n-d-1}.$$

If R contains no nonvoid block, then $\Delta_d^t(G) = B$ from the outset. $\square$

Remark 4.2. Theorem 4.1 does not use the simple connectivity of $\mathrm{BI}_2(G_i)$. It is therefore slightly stronger than the $k = d$ specialization requested in [2, Question 29].

5 The case $k = d + 1$

Write

$$e = n - k = \sum_{i=1}^{k} (|V(G_i)| - 1). \tag{5.1}$$

The target sphere dimension is exactly e.

Theorem 5.1. *Under the hypotheses of Theorem A, if $k = d + 1$, then*

$$\Delta_d^t(G) \simeq \bigvee_d S^{n-d-1}. \tag{5.2}$$

Proof. If $e = 0$, all components are singletons. Then

$$\Delta_d^t(G) = \{\sigma : |\sigma| \leq 1\},$$

which is a set of $d + 1$ points and hence a wedge of d copies of S^0.

If $e = 1$, there is one two-vertex connected component $\{a, b\}$ and d singleton components. The complex is the graph obtained from $K_{2,d}$ by adding the edge ab. It is connected and has cycle rank

$$(2d + 1) - (d + 2) + 1 = d.$$

Thus it is a wedge of d circles.

Assume now that $e \geq 2$. If a component with at least three vertices exists, choose one such component as C; otherwise choose any nonsingleton component as C, in which case every nonsingleton component is a K_2. Fix $a \in C$. For every other vertex x, deleting a and x empties at most one component. By Lemma 2.1, ax is an edge of $\Delta_d^t(G)$. Hence the star with center a is a spanning tree of the 1-skeleton. The fundamental group is generated by loops

$$a - x - y - a, \tag{5.3}$$

where xy is an edge not in the star.

First suppose $|C| \geq 3$. Choose a neighbor b of a in the connected graph C. If axy is a face, the loop (5.3) is filled. Otherwise $b \notin \{x, y\}$. The triples abx and aby are faces, since deleting either triple empties at most one component. Because xy is an edge, there is an independent d-set S in $G - \{x, y\}$. The failure of axy to be a face forces $a \in S$. Since ab is an edge of G, we have $b \notin S$, so the same S witnesses that bxy is a face. The triangles abx, aby, and bxy form a disk with boundary (5.3).

It remains to consider the case in which every nonsingleton component is a K_2. Since $e \geq 2$, there are at least two such components. Write one as $\{a, b\}$ and choose a vertex c in another. If axy is not a face while xy is an edge, then necessarily $\{x, y\} = \{b, s\}$ for a singleton component s. Indeed, if b is absent, failure of ab's star triangle requires x and y to exhaust two singleton components, contradicting that xy is an edge; if b is present and the other point is not a singleton, at most one component is exhausted. In the remaining case, abc, acs, and bcs are faces and form a disk with boundary $a - b - s - a$.

Every generator (5.3) is therefore null-homotopic, so $\Delta_d^t(G)$ is simply connected for $e \geq 2$. Proposition 3.1 says that its only nonzero reduced homology is $\mathbb{Z}^d$ in dimension e. The standard wedge-of-spheres criterion used in [2, Theorem 5] now gives (5.2). □

6 The case $k = d + 2$

Theorem 6.1. *Under the hypotheses of Theorem A, if $k = d + 2$, then*

$$\Delta_d^t(G) \simeq \bigvee_{\binom{d+1}{2}} S^{n-d-1}. \tag{6.1}$$

Proof. Deleting any two vertices empties at most two components, so at least d nonempty components remain. Lemma 2.1 shows that the 1-skeleton of $\Delta_d^t(G)$ is complete.

Suppose some component C has at least two vertices, and fix $v \in C$. For any two other vertices x, y, the triple vxy cannot exhaust three components: exhausting a component with v requires another vertex of C, leaving only one vertex available to exhaust another component. Hence vxy is a face. Every loop in the complete 1-skeleton is filled by a triangular fan with apex v. Thus the complex is simply connected. Here $n \geq d + 3$, so the target dimension $n - d - 1$ is at least 2. Proposition 3.1 and the wedge-of-spheres criterion give (6.1).

If every component is a singleton, then $G = (d + 2)K_1$ and

$$\Delta_d^t(G) = \{\sigma : |\sigma| \leq 2\} = \mathrm{sk}_1 \Delta^{d+1}.$$

This is the complete graph on $d + 2$ vertices. Its cycle rank is

$$\binom{d+2}{2} - (d+2) + 1 = \binom{d+1}{2},$$

which again gives (6.1). □

7 Completion of the disconnected formula

Proof of Theorem A. The cases $k = d$, $k = d + 1$, and $k = d + 2$ are Theorems 4.1, 5.1, and 6.1. If $d \geq 3$, the range $k \geq d + 3$ is [2, Theorem 28].

It remains only to make the $d = 2$, $k \geq 5$ range independent of the parameter restriction in [2, Theorem 26]. Proposition 3.1 gives

$$\widetilde{H}_{n-3}(\Delta_2^t(G); \mathbb{Z}) \cong \mathbb{Z}^{k-1}$$

and no other reduced homology. Deleting any three vertices empties at most three components, so at least $k - 3 \geq 2$ nonempty components remain. Lemma 2.1 shows that the 2-skeleton of $\Delta_2^t(G)$ is the full 2-skeleton on its vertex set. Hence the complex is simply connected. Since $n - 3 \geq 2$, the wedge-of-spheres criterion [2, Theorem 5] gives

$$\Delta_2^t(G) \simeq \bigvee_{k-1} S^{n-3}.$$

These ranges exhaust all $d \geq 2$ and $k \geq d$. □

The binomial multiplicities agree on the three boundary layers:

$$\binom{d-1}{d-1} = 1, \qquad \binom{d}{d-1} = d, \qquad \binom{d+1}{d-1} = \binom{d+1}{2}.$$

The proofs also explain why the single complete-2-skeleton argument of [2, Theorem 28] does not simply extend downward. At $k = d + 2$ only selected triangles are needed; at $k = d + 1$ one must first choose a spanning star and fill its fundamental cycles; at $k = d$ the correct object is the entire weak-composition cover rather than the 2-skeleton.

8 Relation to previous decomposition methods

The proof uses several established tools whose scope should be separated from the new conclusion.

First, for $d \geq 3$, the homology input is inherited from the bounded-independence calculation and Alexander duality in [2, Theorems 26 and 28]. For $d = 2$, Proposition 3.1 supplies the elementary clique-component/Alexander-duality argument directly. Second, the passage from local equivalences on a basis-like cover to a global weak equivalence is [2, Theorem 10]. Third, [2, Theorem 34] gives a two-graph decomposition of $\Delta_d^t(G_1 \sqcup G_2)$ when the relevant filtration inclusions are null-homotopic. This already identifies independent-number splitting as a natural mechanism for disjoint unions.

Theorem 4.1 is not a direct iteration of that two-factor result. After combining several components, the intermediate total-cut filtration need not retain the componentwise contractibility and null-homotopy hypotheses. The weak-composition cover instead handles all distributions simultaneously and compares the full nonbaseline diagram to its trace on the all-ones block.

Polyhedral joins and their homotopy-colimit decompositions are also established objects; see [3]. We do not claim a new general theorem about polyhedral joins. The retained point is the total-cut-specific cover and its consequence for the three component counts asked about in [2, Question 29].

9 Boundary of the result

Theorem A is conditional on the same component assumptions as [2, Theorem 28]. It does not classify $\Delta_d^t(G)$ for arbitrary disconnected graphs. In particular, it does not answer [2, Question 30], which asks whether the clique-complex connectivity condition can be removed.

The $k = d$ argument shows that one hypothesis can be removed in that single layer, but it does not imply a corresponding weakening for $k > d$. Nor does the proof determine simple-homotopy type, shellability, torsion outside the stated hypotheses, or the maps between consecutive total-cut filtration levels.

No computer enumeration is used in the proof. Finite calculations performed during discovery served only as falsification controls and are not evidence for the universal statements.

10 AI-assisted research and writing disclosure

This work was developed with AI assistance for literature discovery, proof stress-testing, symbolic organization, and drafting. Carptopus is the responsible author and has reviewed the mathematical claims and the disclosed computational boundaries. OpenAI Codex is a research and writing tool, not an author.

References

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